

GLOSSARY

Abyssal plain A flat area on the floor of the deep ocean, beyond the continental shelf, at a depth of 13,000–20,000 ft (4,000–6,000 m).

Algae Plantlike organisms that make sugar using the energy of sunlight. Seaweeds are large algae.

Atmospheric pressure The pressure created by the weight of air in the atmosphere, normally measured at sea level.

Atoll A ring-shaped island formed from a coral reef based on a sunken extinct volcano.

Bacteria Microscopic organisms with a simple single-celled structure. Some types can make food using energy that they get from sunlight or chemical reactions.

Barrier reef A coral reef that protects a shallow lagoon from the deep ocean.

Basalt A dark, heavy volcanic rock that forms oceanic crust and erupts as molten lava from midocean ridges and hotspot volcanoes.

Bedrock The solid rock that lies beneath more recent, softer sediments.

Biogenic ooze A soft sediment formed from the skeletal remains of microorganisms such as plankton.

Black smoker A hot spring or hydrothermal vent on the ocean floor, usually at a midocean ridge, that erupts dark, cloudy, mineral-rich water.

Bycatch Animals such as the wrong type of fish, marine mammals, and seabirds that are accidentally caught when fishing.

Carbohydrates Compounds of carbon, hydrogen, and oxygen that store energy, made by some living things and used as food. Sugars are simple carbohydrates.

Carbon dioxide A gas that forms a very small percentage of the atmosphere. Living things such as plants and phytoplankton use it to make food, and it is also a greenhouse gas.

Chemosynthesis Using the energy of chemical reactions to make food (carbohydrates) from carbon dioxide and water.

Chlorophyll A substance that absorbs the energy of sunlight, used by some living things to make food in the process of photosynthesis.

Coccolith The skeleton of a microscopic marine organism called a coccolithophore, which in large numbers forms limestone or chalk rock.

Condense To turn from a gas to a liquid.

Continental crust A thick slab of relatively light rock that "floats" on the Earth's mantle.

Continental rise The slope that links the edge of the continental slope with the deep ocean floor.

Continental shelf The submerged fringe of a continent, which lies beneath a coastal sea.

Continental slope The edge of the continental shelf that slopes down to the continental rise and ocean floor.

Convection The movement and circulation of gases and liquids in response to heat.

Convergent boundary A boundary between two plates of the Earth's crust that are moving together, marked by earthquakes and volcanoes.

Corals Animals related to sea anemones that often form reef-building colonies.

Crustacean An animal with a hard, shell-like external skeleton and paired, jointed legs, such as a crab or shrimp.

Current A flow of ocean water, driven by the wind or by differences in water density caused by temperature and salt content.

Cyclone A weather system marked by clouds, rain, and strong winds, caused by air swirling into a region of rising warm, moist air.

Dark zone The deep region of the ocean where there is no light.

Decompose To rot.

Delta An accumulation of sand and silt laid down at the mouth of a river, usually with several tributary channels flowing over it.

Density The compactness of a substance. If the substance is squeezed together, it becomes more dense.

Depression An area of low atmospheric pressure in which warm air converges and rises. Also known as a cyclone.

Divergent boundary A boundary between two plates of the Earth's crust that are moving apart.

DNA (Deoxyribonucleic acid) The complex protein-like substance that contains the instructions needed to direct the growth of a living thing.

Downwelling zone A sea area where water is sinking.

Echo-sounding Using sound pulses (sonar) to measure water depth, or detect schools of fish.

Ekman spiral The way moving water swerves increasingly to the right or left with depth, so it moves in a different direction from the surface water.

El Niño A change in the ocean currents of the equatorial Pacific, when warm surface water moves east to suppress the normal flow of colder water. This affects the oceanic food supply and weather.

Erosion Wearing away, usually by natural forces such as waves on the shore.

Estuary A river mouth.

Evaporate To turn from a liquid to a gas.

Food web The complex relationship between living things that feed on each other.

Fossil fuel A carbon compound that can be burned to release energy, such as coal, oil, or natural gas, made by the decomposition of dead organisms over millions of years.

Fringing reef A coral reef that lies along a rocky shoreline (continent or island), and does not enclose a well-developed shallow lagoon.

Geyser A jet of hot water and steam that regularly erupts from volcanically heated rocks.

Glacier A mass of ice that is flowing very slowly downhill, usually through a deep valley.

Global conveyor The linked system of currents that carries ocean water around the globe.

Granite A crystalline, hard rock that is one of the main rocks found in continental crust.

Greenhouse effect The warming effect caused by the way atmospheric gases such as carbon dioxide, methane, and water vapor absorb some of the heat radiated from Earth, and stop it from escaping into space.

Gyre A large-scale circular pattern of ocean currents, rotating clockwise north of the equator, and counterclockwise south of the equator.

Headland A narrow area of coastal land projecting between two bays.

Hotspot An unusually hot part of the Earth's thick mantle, which makes volcanoes erupt through the crust above.

Hydrothermal vent A hot spring in the ocean floor, normally on a spreading rift. Also known as a black smoker.

Intertidal Refers to the area of the shore that is covered and uncovered by the tides.

Invertebrate An animal that does not have an internal skeleton based on spinal bones.

Island arc A line of islands marking a boundary between two plates of the Earth's crust, created by volcanic activity as one plate plunges beneath the other and is destroyed.

Krill Oceanic shrimp that form large swarms in the Southern Ocean, and are the main food of most Antarctic animals.

Lagoon An area of shallow water that has been cut off from the sea.

Lava Molten rock that erupts from volcanoes or volcanic fissures.

Longline A very long fishing line equipped with thousands of baited hooks.

Longshore drift The movement of beach material along the shore by waves.

Magma Molten rock that has not erupted and is still contained within the Earth's crust.